

Martian Atmospheric Dust Abrasion & Electrostatic Repulsion

1. Triboelectric Charge Characteristics

Atmospheric suspension imparts strong negative triboelectric charges to iron oxide dust grains, causing high adhesive electrostatic affinity to photovoltaic glass and suit fabrics.

2. Electrostatic Dust Shielding (EDS)

Solar arrays feature embedded indium tin oxide electrode grids pulsing high-voltage three-phase AC electric fields to levitate and sweep dust grains off panel surfaces.

3. Rotary Joint Labyrinth Seals

Air lock hatches and rover wheel bearing assemblies utilize triple magnetic fluid ferrofluidic seals and pressurized nitrogen purge barriers to prevent particulate ingress.